

End-to-End MLOps Pipeline

Sentiment Classification with DistilBERT

Docker · GitHub Actions · Kaggle · W&B
MLOps · PGD AI Program · IIT Jodhpur · Group 1

1. Team Members and Contributions

Every member of the team has contributed equally and worked diligently towards this project.

Team Member	Roll No.	Contribution
Ashwani Bhardwaj	G25AIT2022	Data collection and preprocessing, IMDB dataset preparation, DistilBERT fine-tuning on Kaggle GPU, experiment tracking using Weights & Biases, Hugging Face deployment
Jeenal Chaudhary	G25AIT2027	GitHub Actions workflow development (ci.yml and inference.yml), workflow testing and validation, README.md enhancement, documentation, project report preparation
Rakesh Kumar Panda	G25AIT2082	GitHub repository administration, collaborator management, branch protection configuration, workflow integration, repository governance
Sai Harshith Duggisetty	G25AIT2033	Initial dataset exploration, model selection support, Docker containerization, deployment support, README.md creation

2. Architecture

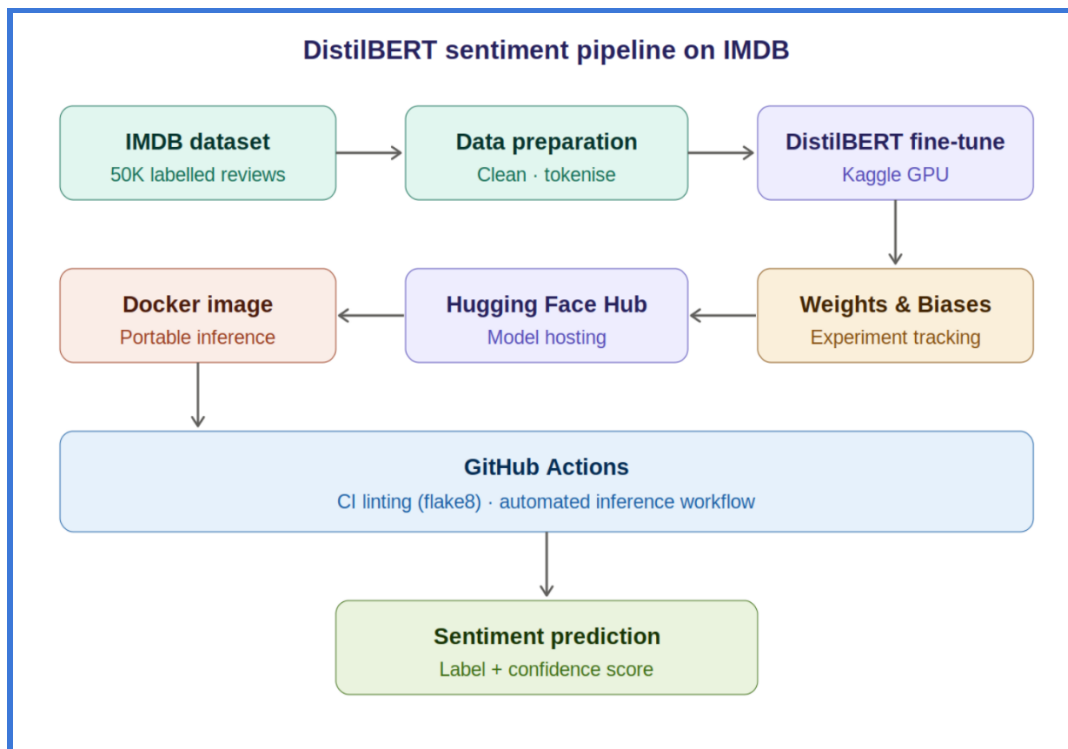
➤ High-Level Design

The project implements a complete MLOps pipeline for sentiment classification using DistilBERT on the IMDB Movie Reviews dataset.

- Pipeline Flow

Dataset (IMDB) → Data Cleaning & Preprocessing → DistilBERT Fine-Tuning (Kaggle GPU) → Experiment Tracking (W&B) → Hugging Face Deployment → Docker Containerization → GitHub Actions Automated Inference

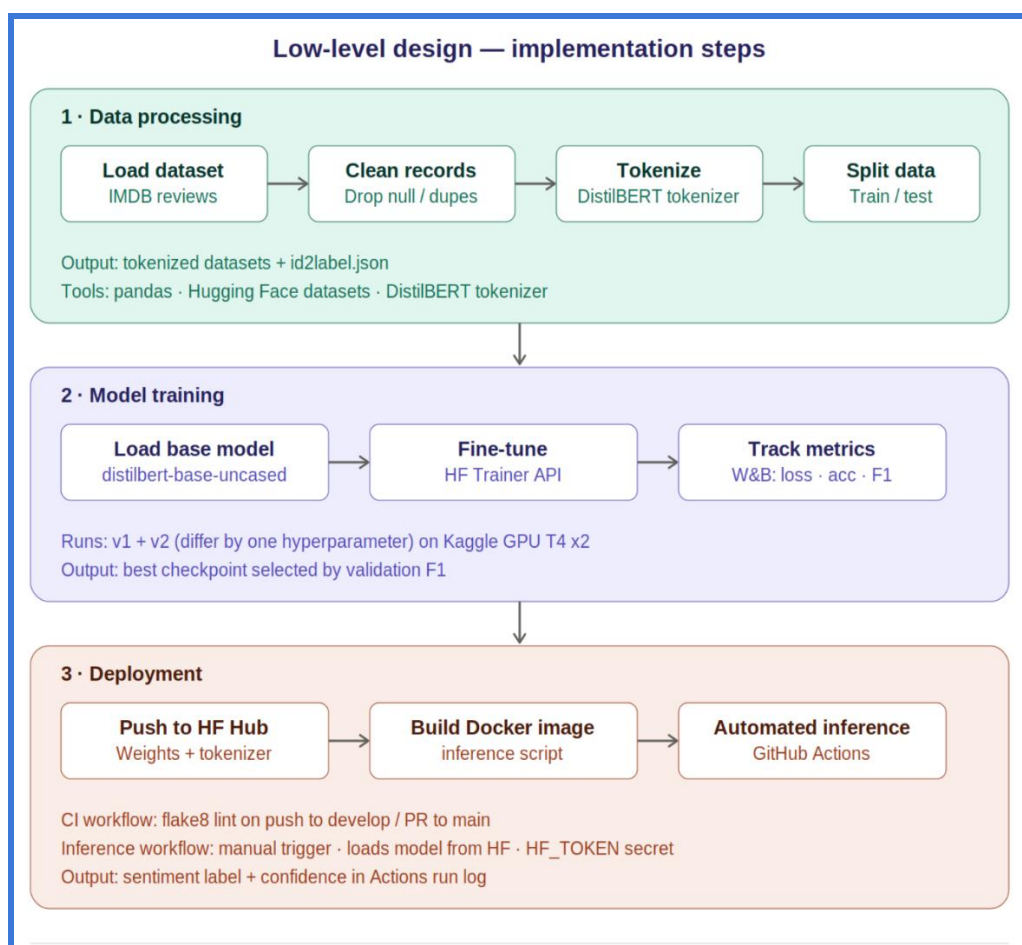
- **Components**
 - GitHub Repository for version control and CI/CD.
 - Kaggle GPU environment for model training.
 - Weights & Biases for experiment tracking.
 - Hugging Face Hub for model hosting.
 - Docker for portable inference deployment.
 - GitHub Actions for automated testing and inference.



➤ Low-Level Design

- **Data Processing**
 - Load IMDB dataset.
 - Remove null and duplicate records.
 - Tokenize reviews using DistilBERT tokenizer.
 - Generate train/test splits.
- **Model Training**
 - Load DistilBERT base uncased model.
 - Fine-tune using Hugging Face Trainer API.
 - Track metrics using W&B.
- **Deployment**
 - Push best-performing model to Hugging Face Hub.
 - Build Docker image using inference script.

- Execute automated inference through GitHub Actions.



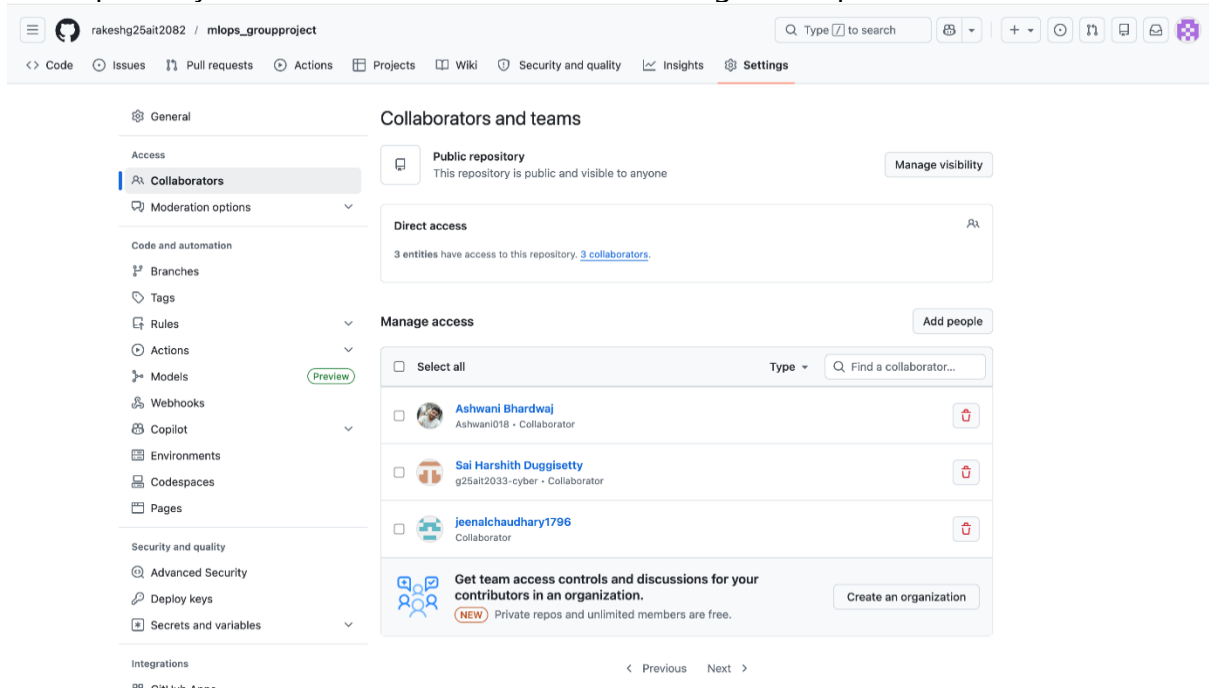
3. Live Links

Component	Link
GitHub Repository	https://github.com/rakeshg25ait2082/mlops_groupproject
Kaggle Notebook – Version 1	https://www.kaggle.com/code/ashwanig20ait2022/mlops-project-transformers-model-imdb-v-1?scriptVersionId=325555945
Kaggle Notebook – Version 2	https://www.kaggle.com/code/ashwanig20ait2022/mlops-project-transformers-model-imdb-v-1?scriptVersionId=325561163
Hugging Face Model	https://huggingface.co/Nlp0187/distilbert-imdb-reviews-v1
Docker Image	https://hub.docker.com/r/ashwani0187/mlops-a3-inference
W&B Dashboard	https://wandb.ai/ashwinmmec794-iit-jodhpur/Mlops-Group-Project

4. Git Repository Setup

Collaborators

The repository contains four contributors with assigned responsibilities.



Branch Strategy

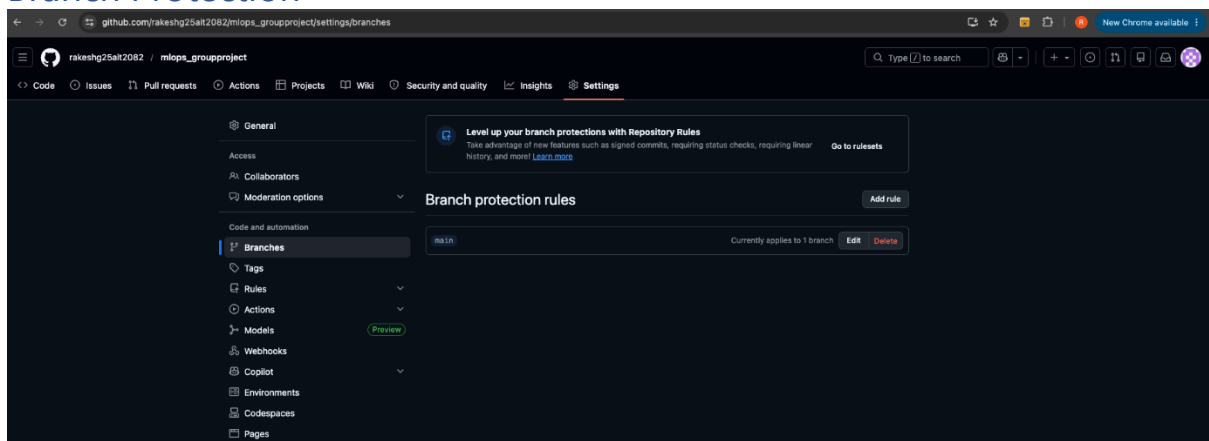
Branches used:

- main
- develop

Branch protection was enabled to:

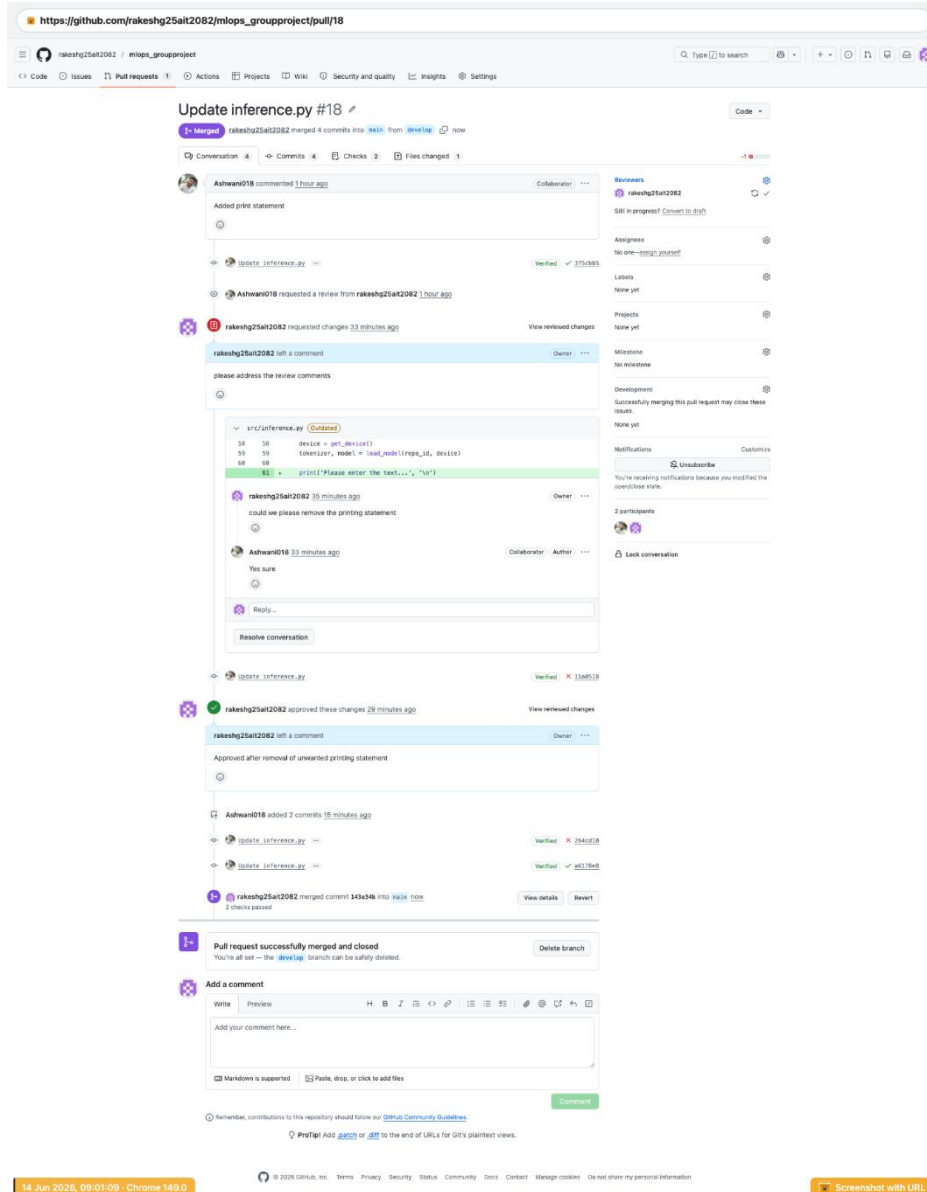
- Requires at least 1 Pull-request review
- Prevent direct pushes
- Ensure workflow validation before merge

Branch Protection



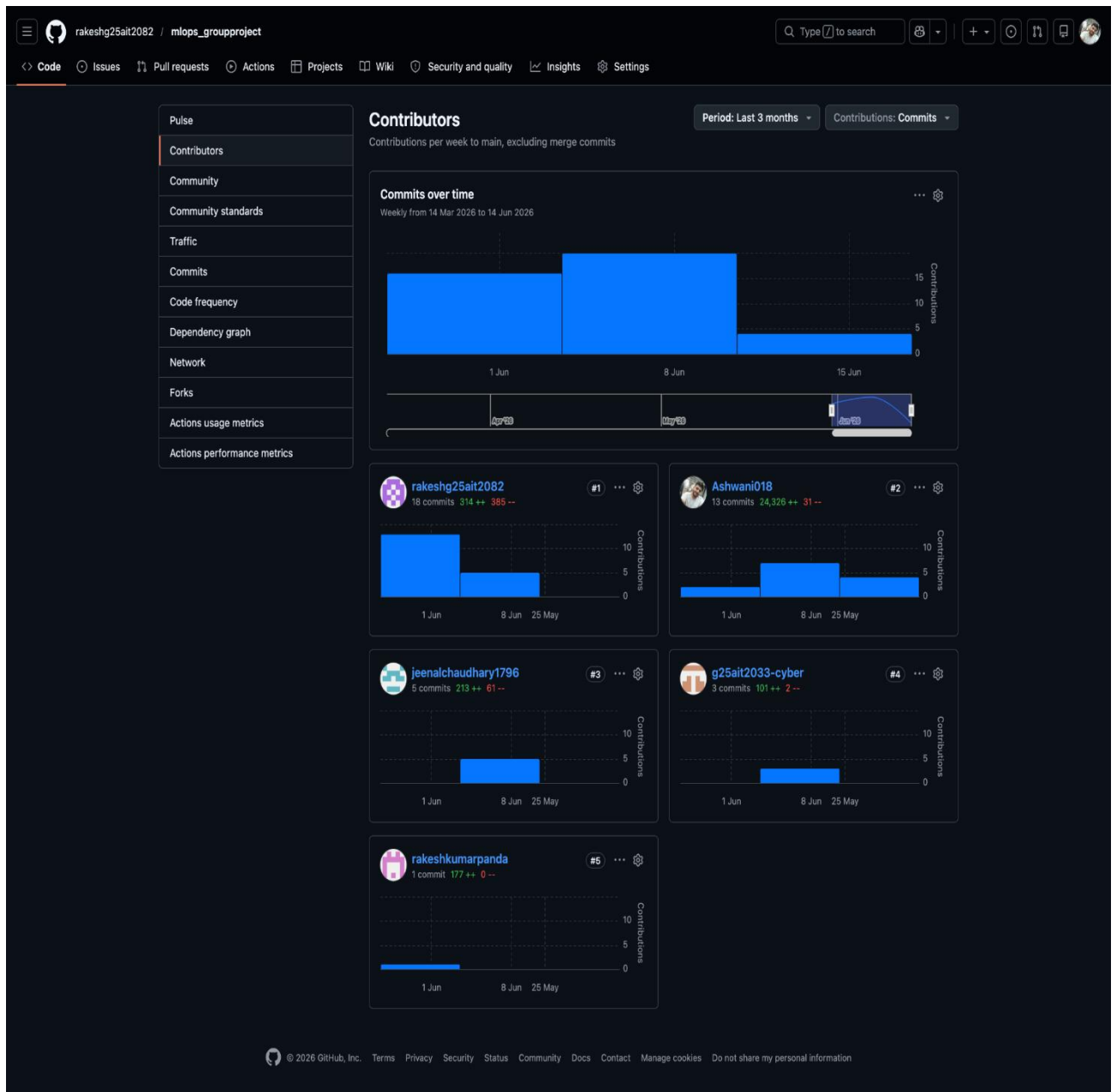
PR Review in Practice

A pull request review process was followed where code changes were reviewed before merging into protected branches.



Contributors History

The GitHub Contributors page confirms contributions from all four members through commits and merged pull requests.



5. Data Cleaning Decisions (Task 2)

Dataset

IMDB Movie Reviews Dataset

Text Cleaning Decisions

HTML Entity Decoding

Converted encoded HTML entities (e.g., `&`, `"`, `'`) into their corresponding characters to improve text readability and preserve semantic meaning.

HTML Tag Removal

Removed HTML tags such as `<br>`, `<p>`, and `<div>` to eliminate formatting artifacts that do not contribute to sentiment classification.

URL Removal

Removed web links and URLs, as they generally do not contain useful sentiment information and may introduce noise into the model.

Whitespace Normalization

Replaced multiple spaces, tabs, and newline characters with a single space to ensure consistent text formatting.

Leading and Trailing Space Removal

Removed unnecessary spaces at the beginning and end of each review to produce clean and standardized input text.

Why Text Cleaning?

These preprocessing steps were applied to remove irrelevant information and formatting artifacts while preserving the original semantic content of the reviews. The approach follows a minimal-cleaning strategy suitable for transformer-based models such as DistilBERT, which are designed to learn contextual representations directly from natural language text.

6. Model Selection Rationale (Task 3)

Selected Model

DistilBERT Base Uncased

Why DistilBERT?

DistilBERT Base Uncased was selected because it provides strong text classification performance while being smaller and faster than BERT. According to the Hugging Face model card, it retains most of BERT's language understanding capabilities with fewer parameters and lower computational requirements.

Its efficient architecture, strong sentiment analysis performance, and seamless integration with the Hugging Face ecosystem made it an ideal choice for building and deploying the end-to-end MLOps pipeline. The final model achieved **92.85% Accuracy** and **92.84% F1 Score** on the IMDb sentiment classification task.

Hugging Face Reference Model Card

Model [1] was loaded directly using

```
AutoTokenizer.from_pretrained("distilbert-base-uncased")  
AutoModelForSequenceClassification.from_pretrained("distilbert-base-uncased")
```

[1] Hugging Face, "DistilBERT Base Uncased," Hugging Face Model Hub.
Available: <https://huggingface.co/distilbert/distilbert-base-uncased>

7. Experiment Comparison (Task 4)

Two training experiments were executed.

Metric	Version 1	Version 2
Learning Rate	3e-5	1e-5
Batch Size	16	16
Epochs	3	3
Accuracy	92.85%	91.70%
F1 Score	92.84%	91.69%

Winner Model

Version 1 achieved the highest Accuracy and F1 Score, indicating better sentiment classification performance on the validation dataset. Although Version 2 achieved a lower validation loss, the lower Accuracy and F1 Score suggest that the model did not learn the decision boundary as effectively within the available three training epochs.

The reduction in learning rate from 3e-5 to 1e-5 resulted in slower convergence. As a consequence, Version 2 may require additional training epochs to reach performance comparable to Version 1.

Based on the experimental results, **Version 1** was selected as the final model for deployment. It achieved the highest Accuracy (92.85%) and F1 Score (92.85%), making it the best-performing model among the two experiments. While Version 2 demonstrated slightly better validation loss, the overall classification performance of Version 1 was superior and therefore more suitable for the sentiment analysis task.





Kaggle Secrets Configured

- HF_TOKEN
- WANDB_API_KEY

mlops-project-transformers-model-imdb-v[1] 8f7... Draft saved

File Edit View Run Settings Add-ons Help

+ Draft Session off (run a cell to start)

```
[ ]: !pip3 install -U transformers
```

```
[ ]: # Basic Python modules
from collections import defaultdict
import random
import pickle

# For downloading large files from Google Drive
# https://github.com/wkentaro/gdown
import gdown

# For working with gzip files
# https://docs.python.org/3/library/gzip.html
import gzip

# For working with JSON files
import json

# For data manipulation and analysis
import pandas as pd
import numpy as np
```

Secrets

Code Snippet

```
from kaggle_secrets import UserSecretsClient
secret_label = "your-secret-label"
secret_value = UserSecretsClient().get_secret_value(secret_label)
```

Copy to clipboard

HF_TOKEN

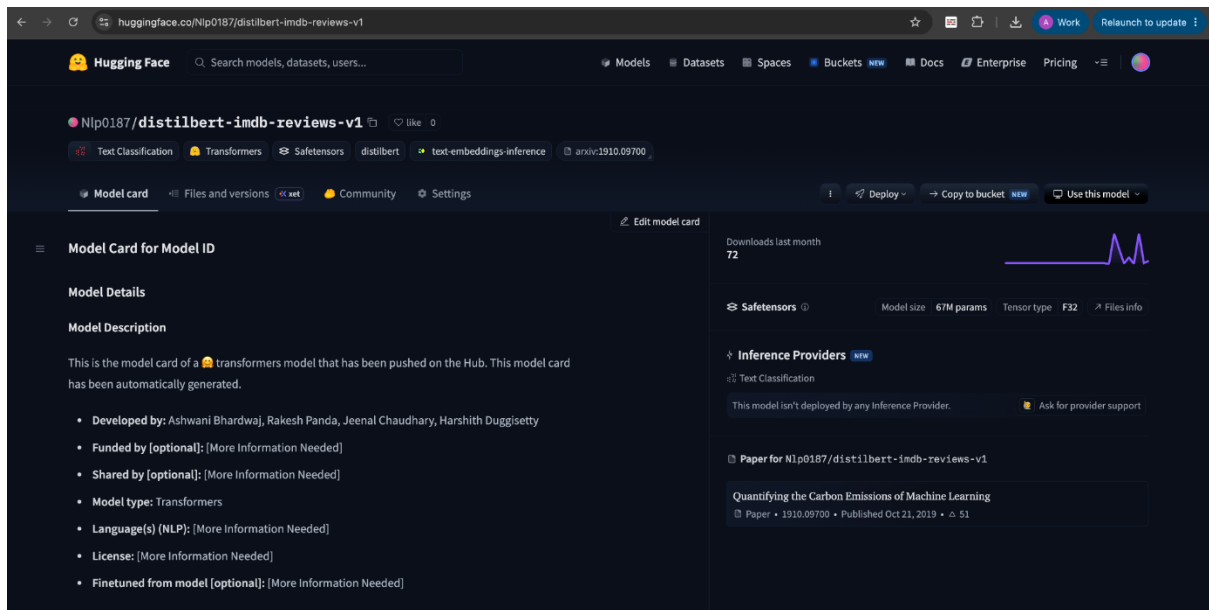
WANDB_API_KEY

+ Add Secret Copy snippet

8. Deployment & Automation (Task 5)

Hugging Face Deployment

The final model was pushed to Hugging Face Hub and made publicly accessible.



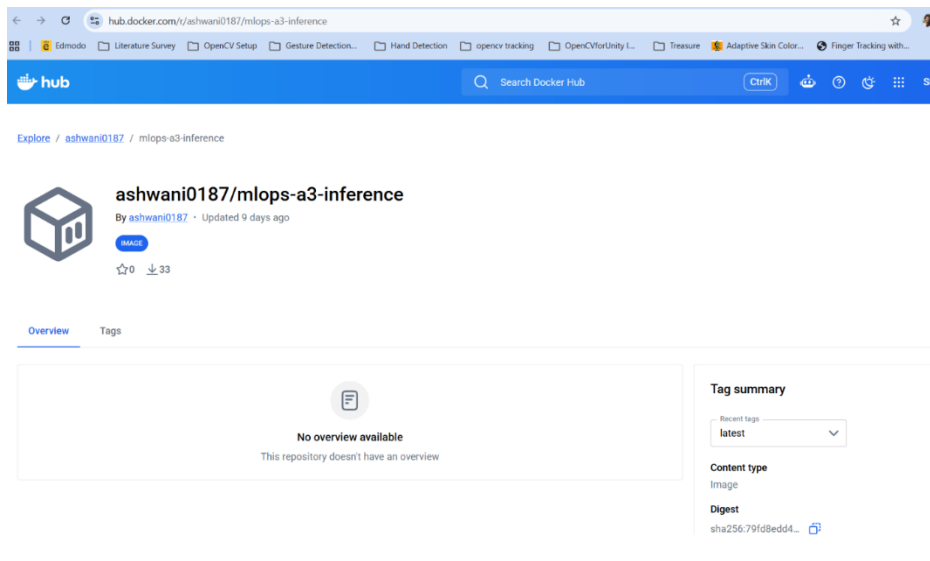
9. Docker Design (Task 6)

Dockerfile Decisions

- Python 3.11 slim base image.
- Lightweight dependencies.
- Cached dependency layers.
- Environment-variable-driven inference.

Docker Image

Public Docker image published for inference.



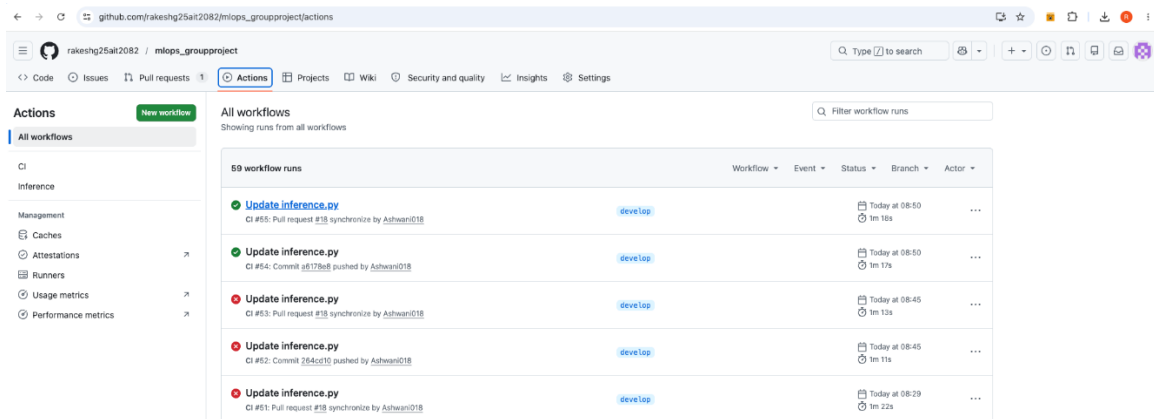
10. GitHub Actions (Task 7)

Two workflows have been implemented:

➤ ci.yml

- Code quality validation
- Workflow testing

ci.yml Run (triggered automatically as soon as there is a new commit)



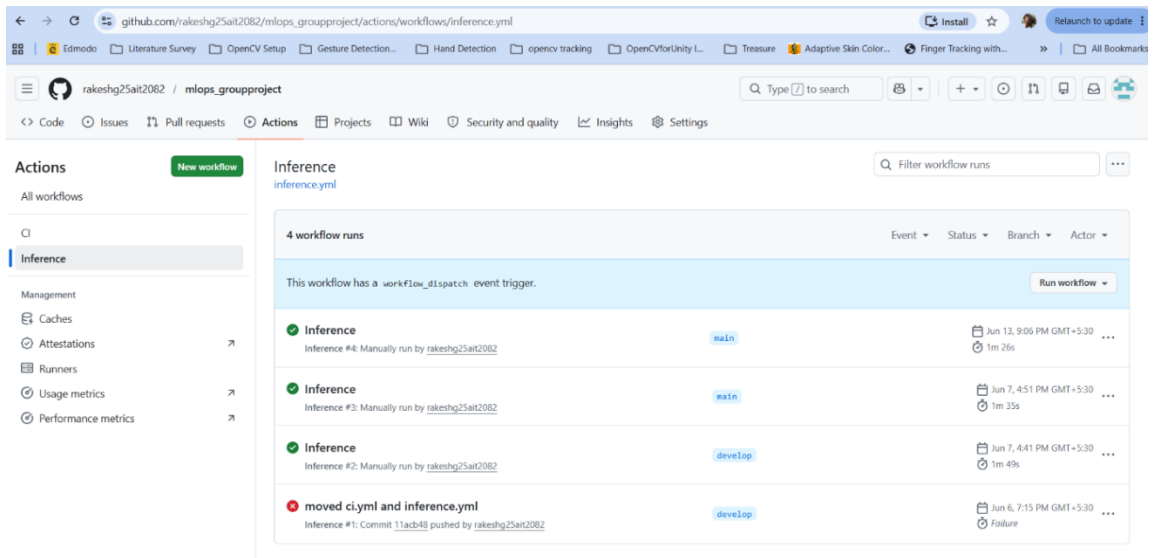
The screenshot shows the GitHub Actions interface for the repository 'rakeshg25ait2082 / mlops_groupproject'. The 'Actions' tab is selected, and the 'All workflows' section is displayed. A table lists 59 workflow runs for the 'Update inference.py' workflow. The table columns are Workflow, Event, Status, Branch, and Actor. The runs are filtered by the 'develop' branch. The runs are triggered by 'Pull request #18 synchronize by Ashwan018' and 'Commit 264d010 pushed by Ashwan018'. The status of the runs is 'Success' (green circle) or 'Failure' (red circle). The time taken for each run is shown in minutes and seconds.

Workflow	Event	Status	Branch	Actor
Update inference.py	CI #55: Pull request #18 synchronize by Ashwan018	Success	develop	Ashwan018
Update inference.py	CI #54: Commit 26178e8 pushed by Ashwan018	Success	develop	Ashwan018
Update inference.py	CI #53: Pull request #18 synchronize by Ashwan018	Success	develop	Ashwan018
Update inference.py	CI #52: Commit 264d010 pushed by Ashwan018	Success	develop	Ashwan018
Update inference.py	CI #51: Pull request #18 synchronize by Ashwan018	Success	develop	Ashwan018

➤ inference.yml

- Manual inference execution
- Hugging Face model loading

Inference Run (triggered manually)



The screenshot shows the GitHub Actions interface for the repository 'rakeshg25ait2082 / mlops_groupproject'. The 'Actions' tab is selected, and the 'Inference' workflow is displayed. A table lists 4 workflow runs for the 'Inference' workflow. The table columns are Event, Status, Branch, and Actor. The runs are triggered by 'Manually run by rakeshg25ait2082' and 'Commit 11acb48 pushed by rakeshg25ait2082'. The status of the runs is 'Success' (green circle) or 'Failure' (red circle). The time taken for each run is shown in minutes and seconds.

Event	Status	Branch	Actor
Inference #4: Manually run by rakeshg25ait2082	Success	main	rakeshg25ait2082
Inference #3: Manually run by rakeshg25ait2082	Success	main	rakeshg25ait2082
Inference #2: Manually run by rakeshg25ait2082	Success	develop	rakeshg25ait2082
moved ci.yml and inference.yml	Failure	develop	rakeshg25ait2082

Inference Run Logs

The screenshot shows the GitHub Actions interface for a workflow named 'infer' in the repository 'rakeshg25ait2082/mllops_groupproject'. The workflow is currently running, and the logs for the 'infer' job are displayed. The logs show the following steps:

- Set up job (8s)
- Run actions/checkout@v4 (1s)
- Run actions/setup-python@v5 (8s)
- Run pip install -r requirements.txt (1m 35s)
- Run python src/inference.py (9s)
- Post Run actions/setup-python@v5 (8s)
- Post Run actions/checkout@v4 (8s)
- Complete job (8s)

The logs for the 'Run python src/inference.py' step show the following output:

```
1 Run python src/inference.py
13 [inference] Loading model: Nlp0187/distilbert-imdb-reviews-v1 on cpu
14
15 Loading weights: 0% | 0/184 [00:00:00, 71it/s]
16 Loading weights: 100% | 184/184 [00:00:00, 3604.18it/s]
17
18 ==> Prediction ==>
19 Input : This is a nice movie
20 Output: {
21   "label": "POSITIVE",
22   "score": 0.9957
23 }
```

GitHub Secrets

The screenshot shows the GitHub repository settings for 'rakeshg25ait2082/mllops_groupproject'. The 'Secrets and variables' section is selected, and the 'Actions secrets and variables' tab is active. The page displays the following information:

- Environment secrets:** This environment has no secrets. A button 'Manage environment secrets' is available.
- Repository secrets:** A table showing the repository secrets:

Name	Last updated
HF_TOKEN	last week
WANDB_API_KEY	last week

11. Show All Experiments on W&B (Task 8)

All experiments were tracked using Weights & Biases (W&B) to monitor the training process and compare model configurations. Two training runs were logged under the same W&B project, with each run recording training and validation metrics across

epochs. The project visibility was set to Public to allow access to the experiment results. The W&B Runs Comparison feature was used to display the key metrics side-by-side, enabling a clear comparison between the two model configurations. A screenshot of the W&B dashboard and the project URL are included to provide evidence of experiment tracking and reproducibility.

WandB Project URL:

<https://wandb.ai/ashwinmmec794-iit-jodhpur/Mlops-Group-Project>

Comparison Table of Multiple Model Versions:

Metric	Version 1	Version 2
Training Loss	0.15	0.2962
Validation Loss	0.5502	0.4552
Accuracy	92.85%	91.70%
F1 Score	92.84%	91.69%

12. Challenges and Learnings

One of the main challenges during the project was integrating multiple MLOps components into a single reproducible workflow. Managing GitHub collaboration, branch protection rules, pull request reviews, GitHub Actions workflows, Hugging Face deployment, Docker containerization, and Weights & Biases tracking required careful coordination among team members.

Another challenge was configuring secure authentication for external services such as Hugging Face and W&B using Kaggle Secrets and GitHub Secrets.

In this project, we faced the major issue of merging to main when we configured ruleset to protect the main branch, which we overcame by reconfiguring to use classic branch protection. Also, a couple of CI runs failed, which we fixed by debugging the errors.

Through this project work, as a part of learning, the team gained practical experience in end-to-end MLOps practices, including experiment tracking, model versioning, CI/CD automation, collaborative development, and cloud-based model deployment. The project highlighted the importance of reproducibility, automation, and proper repository management in real-world machine learning systems.

What could be done differently?

If we were to repeat the project, we would explore a more production-oriented MLOps setup by incorporating cloud platforms such as AWS and experiment tracking tools such as MLflow. While Weights & Biases and Hugging Face provided an effective

workflow for experiment tracking and deployment, using AWS and MLflow would offer additional experience with cloud-based infrastructure, model registries, artifact management, and scalable deployment pipelines. This would help us better understand how machine learning systems are managed in real-world production environments and broaden our exposure to industry-standard MLOps practices.

13. Conclusion

This project successfully developed and deployed an end-to-end MLOps pipeline for IMDb sentiment classification using DistilBERT. The workflow included data preprocessing, model training on Kaggle, experiment tracking with Weights & Biases, model deployment to Hugging Face Hub, Docker-based packaging, and automated inference using GitHub Actions.

The final model achieved strong classification performance with an Accuracy and F1 Score of 92.85%. Additionally, the project demonstrated key MLOps principles such as collaboration through GitHub, version control, CI/CD automation, experiment management, and reproducible deployments. Overall, the project provided valuable hands-on experience in building production-oriented machine learning systems.